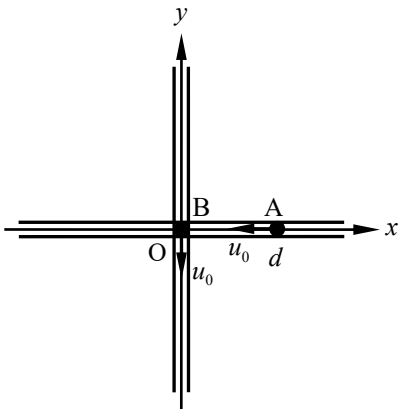
III. (20 points) Two smooth thin tubes intersect at right angles in a level plane; each holds a particle of mass m and charge q.



For A and B, the initial separation of A from crossing O is d; B is at O. Their velocities are perpendicular, both $u_0 = \sqrt{\dfrac{kq^2}{md}}$, as shown; k is Coulomb's constant.

Find their minimum separation in the subsequent motion.